

CO1-AT2-Data Interpretation Exercise

A Roja [192424226]

DSA0601- Data Handling and Visualization

Data Interpretation Assessment (R Programming)

Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

Question 1: Mapping Data onto Aesthetics

Dataset

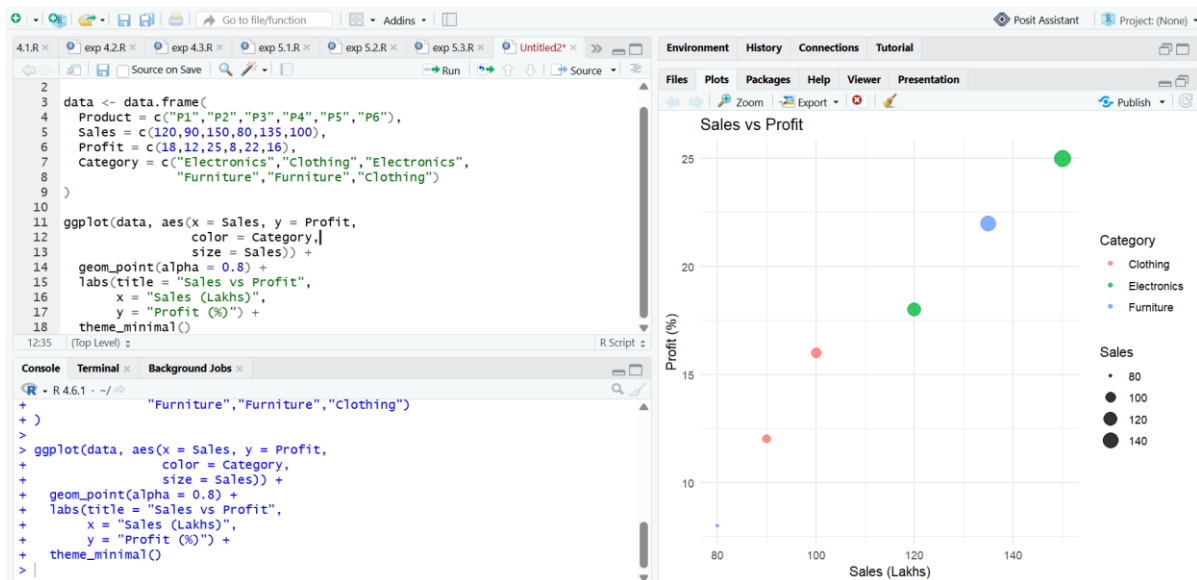
Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

1 Write an R program to create a scatter plot with Sales on the x-axis and Profit on the y-axis.

CODE:

```
library(ggplot2)
data <- data.frame(
  Product = c("P1", "P2", "P3", "P4", "P5", "P6"),
  Sales = c(120, 90, 150, 80, 135, 100),
  Profit = c(18, 12, 25, 8, 22, 16),
  Category = c("Electronics", "Clothing", "Electronics",
               "Furniture", "Furniture", "Clothing")
)
ggplot(data, aes(x = Sales, y = Profit,
                 color = Category,
                 size = Sales)) +
  geom_point(alpha = 0.8) +
  labs(title = "Sales vs Profit",
```

```
x = "Sales (Lakhs)",
y = "Profit (%)" +
theme_minimal()
```



2 Map Category using an appropriate aesthetic and Sales using another suitable aesthetic.

- Category → Color, Sales → Point Size

3 Interpret the relationship between Sales and Profit.

- Higher sales generally correspond to higher profit, indicating a positive relationship.

4 Identify the type of scale used for each variable.

- Sales – Continuous scale
- Profit – Continuous scale
- Category – Categorical (Discrete) scale

5 Suggest one alternative visualization that better represents the same data.

- Bubble Chart

Question 2: Coordinate Systems (Cartesian and Polar)

Dataset

Direction	Frequency
N	18

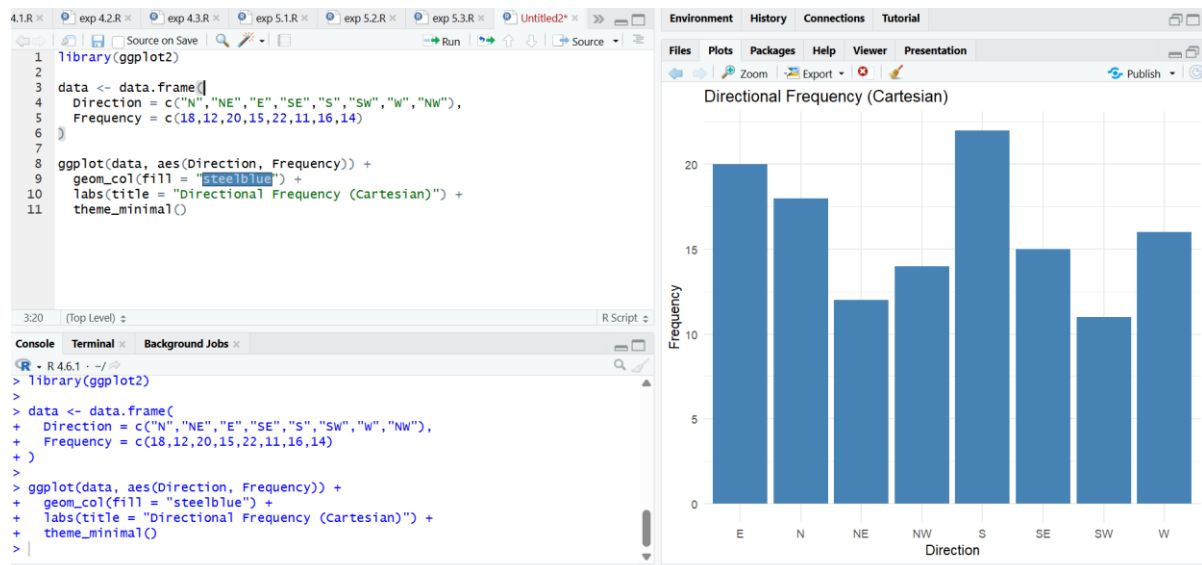
NE	12
E	20
SE	15
S	
SW	22
W	11
NW	16
	14

Questions

1. Write an R program to visualize the dataset using a Cartesian coordinate system.

Cartesian Coordinate System Code

```
library(ggplot2)
data <- data.frame(
  Direction = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
  Frequency = c(18, 12, 20, 15, 22, 11, 16, 14)
)
ggplot(data, aes(Direction, Frequency)) +
  geom_col(fill = "steelblue") +
  labs(title = "Directional Frequency (Cartesian)") +
  theme_minimal()
```



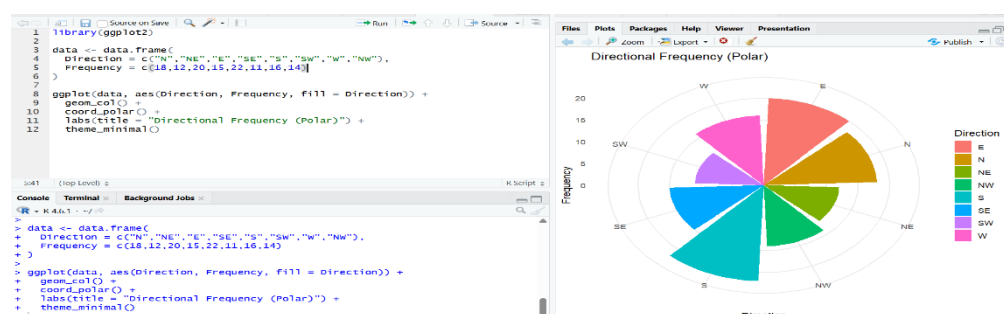
2. Write another R program to visualize the same data using a Polar coordinate system.

Polar Coordinate System Code

```

library(ggplot2)
data <- data.frame(
  Direction = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
  Frequency = c(18, 12, 20, 15, 22, 11, 16, 14)
)
ggplot(data, aes(Direction, Frequency, fill = Direction)) +
  geom_col() +
  coord_polar() +
  labs(title = "Directional Frequency (Polar)") +
  theme_minimal()

```



3 Compare the effectiveness of both visualizations.

- Cartesian is easier for comparing values, while Polar better represents directions.

4 Explain which coordinate system is more appropriate for directional data.

- Polar coordinate system.

5 Justify your answer based on data interpretation.

- Since the data represents compass directions, the circular layout matches the natural orientation of directional data.

Question 3: Color Scales for Data Representation Dataset

Student	Marks
A	45
B	58
C	72

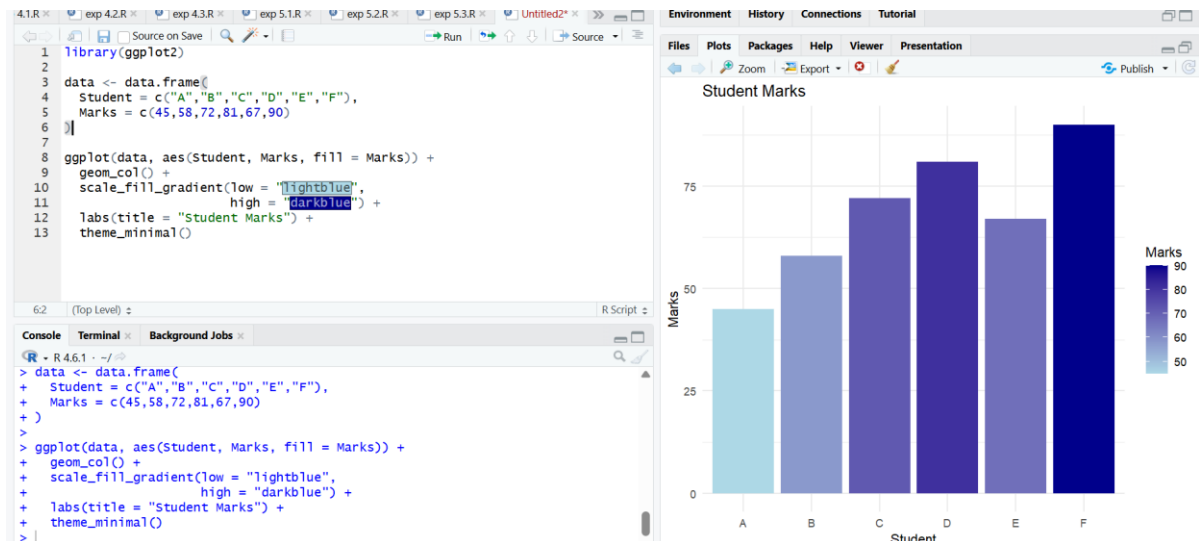
Student	Marks
D	81
E	67
F	90

Questions

- 1. Write an R program to create a bar chart using a sequential color scale based on Marks.**

CODE:

```
library(ggplot2)
data <- data.frame(
  Student = c("A", "B", "C", "D", "E", "F"),
  Marks = c(45, 58, 72, 81, 67, 90)
)
ggplot(data, aes(Student, Marks, fill = Marks)) +
  geom_col() +
  scale_fill_gradient(low = "lightblue",
                     high = "darkblue") +
  labs(title = "Student Marks") +
  theme_minimal()
```



2. Identify the students with the high
 - Highest: Student F (90)
 - Lowest: Student A (45)
3. Explain why a sequential color scale is appropriate.
 - Sequential color scales represent increasing numerical values clearly.
4. Suggest a color palette suitable for color-blind users.
 - Viridis palette.
5. Explain the difference between sequential and categorical color scales.
 - Sequential Scale → Used for continuous numeric values.
 - Categorical Scale → Used for different categories.

Question 4: Highlighting with Color

Dataset

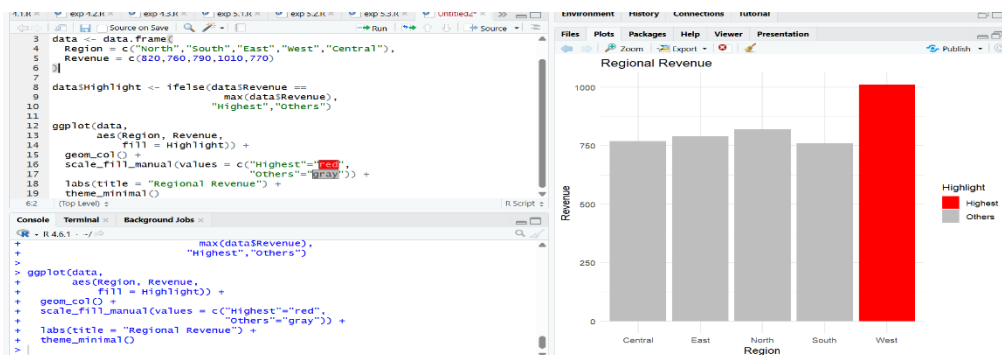
Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

Questions

1. Write an R program to create a bar chart highlighting the region with the highest revenue.

CODE:

```
library(ggplot2)
data <- data.frame(
  Region = c("North", "South", "East", "West", "Central"),
  Revenue = c(820, 760, 790, 1010, 770)
)
data$Highlight <- ifelse(data$Revenue ==
  max(data$Revenue),
  "Highest", "Others")
ggplot(data,
  aes(Region, Revenue,
    fill = Highlight)) +
  geom_col() +
  scale_fill_manual(values = c("Highest"="red",
    "Others"="gray")) +
  labs(title = "Regional Revenue") +
  theme_minimal()
```



2. Explain how highlighting improves data interpretation.
 - Highlighting immediately draws attention to the most important value.
3. Identify the highest-performing region.
 - West Region (₹1010 Crores)
4. Discuss how excessive use of colors can affect interpretation.
 - Too many colors can confuse users and reduce readability.
5. Suggest one alternative highlighting technique other than color.
 - Use labels or annotations.

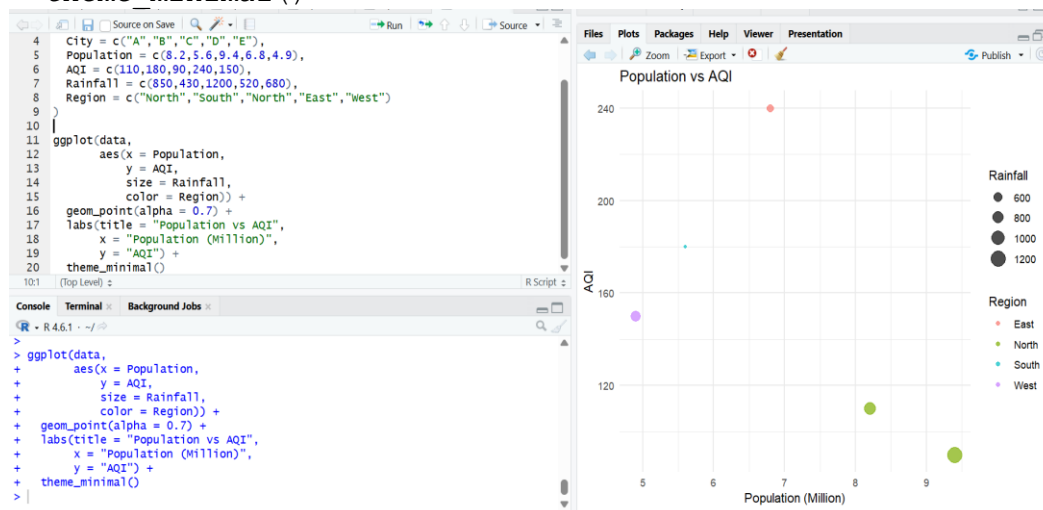
Question 5: Integrated visualization dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6	180	430	South
C	9.4	90	1200	North
D	6.8	240	520	East
E	4.9	150	680	West

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

CODE:

```
library(ggplot2)
data <- data.frame(
  City = c("A", "B", "C", "D", "E"),
  Population = c(8.2, 5.6, 9.4, 6.8, 4.9),
  AQI = c(110, 180, 90, 240, 150),
  Rainfall = c(850, 430, 1200, 520, 680),
  Region = c("North", "South", "North", "East", "West")
)
ggplot(data,
  aes(x = Population,
      y = AQI,
      size = Rainfall,
      color = Region)) +
  geom_point(alpha = 0.7) +
  labs(title = "Population vs AQI",
       x = "Population (Million)",
       y = "AQI") +
  theme_minimal()
```



2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

- Population → X-axis
- AQI → Y-axis
- Rainfall → Bubble Size
- Region → Color

3. Interpret the relationship between Population and AQI.

- Cities with higher population generally tend to have higher AQI, though the relationship is not perfectly linear.

4. Identify the continuous and categorical variables.

- Continuous Variables
 - Population
 - AQI
 - Rainfall
- Categorical Variable
 - Region

5. Recommend one improvement to enhance the visualization's readability.

- Add data labels to identify each city and improve readability.